



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

PREFLASH AND FLASH COLUMN CONDENSER (E-2301)

Docu. No. : 14780-8120-PS-E2301

Project No. : 120117

Rev. No. : Z1

Date : MAY 10, 2016

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TITLE :

PREFLASH AND FLASH COLUMN CONDENSER

(E-2301)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-E2301
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT		S.W.Park	J.S.Chung	- S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE		S.W.Park	J.S.Chung	- S.W.Lee
A1	OCT 8, 2013	APPROVED FOR DESIGN		S.W.Park	J.S.Chung	- S.W.Lee
O1	JUN 27, 2013	COMMENT ISSUE		S.W.Park	J.S.Chung	- S.W.Lee
REV. NO.	DATE	DESCRIPTION		PREP'N	REVIEW	QA REVIEW APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW		APPROVAL
Z1	PROCESS	S.W.Park	J.S.Chung	-		S.W.Lee
		MAY 10, 2016	MAY 10, 2016	-		MAY 10, 2016

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SPECIALTY EXCHANGERS

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GENERAL NOTES

A. Flange Ratings

The flange classes specified are in accordance with ASME B16.5 Latest Edition. For carbon steel flanges the ratings are based on material group 1.1 and for C-½Mo flanges the ratings are based on material group 1.5, unless specified otherwise on the individual specification sheet. Flange material supplied must conform to the requirements and limitations of the material rating specified.

B. Physical Properties

The specified total liquid properties reflect the average properties of all liquid phases.

Each stream may have a different reference basis for the calculation of enthalpies. The normalized enthalpies should not be applied to any other stream or service.

When isobaric properties are provided for an item, the range of duties, temperatures and pressures exceed the terminal conditions of the exchanger to allow for interpolation of all necessary values.

Gauge pressures are based on an atmospheric pressure of 1.0304 kg/cm² [a]

C. Industry Standards and Specifications

The following standards and specifications shall be considered part of this specification:

ASME Boiler & Pressure Vessel Code Section VIII, Division 1

API Standard 662 Part 1 / ISO 15547-1 Plate & Frame Heat Exchangers

D. Additional Notes

1. Supplier (Mandatory):

Ziepack

14 rue de Bassano

75116 Paris

France

Phone: 33 1 53 67 41 41

Fax: 33 1 53 67 41 42

E-mail: france.info@alfalaval.com

www.alfalaval.com

2. A welded plate bundle shall be provided in a pressure containing shell, both sized to meet the process conditions on Sheets 7-10. Process vapor will enter nozzles at the top of the horizontal shell, pass through the welded plate bundle where it cools and condenses. Vapor will exit the vessel through nozzles on the sides, and liquid will exit the vessel through a liquid boot at the bottom of the vessel.

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GENERAL NOTES-continued

3. The maximum allowable pressure drop for the hotside shall reflect the total pressure drop from the vapor inlet nozzle flange, through the exchanger, to the vapor outlet nozzle flange.
4. The condenser design shall provide liquid-vapor separation volume designed to separate liquid droplets greater than 250 microns from the vapor phase of the outlet flow.
5. A liquid boot shall be provided that is sized to provide 4 minute surge time on the total liquid outlet flow. The liquid boot shall have a 2700 mm inside diameter with a length of 3000 mm from the top to the bottom tangent line.
6. Level control nozzles on the liquid boot shall be NPS 2, CL 300 RF flanged connections. Two connections are required at each level (for level gauge and level transmitter). Lower level nozzles shall be located 150 mm above the bottom tangent line of the liquid boot. Upper level nozzles shall be located 350 mm below the top of the liquid boot.
7. Welded connections between the bundle and shell shall be provided where needed to keep process inlet and outlet flows to and from the bundle separate.
8. Vendor to confirm that welded connections between the bundle and the shell (example: a vapor baffle inlet connection is fully seal welded to the top of the bundle and to the shell) can withstand the additional differential pressure that results from a static head of liquid equal to the full height of the bundle.
9. The design shall accommodate differential thermal expansions of the welded plate bundle, frame, support structure, and all internal attachments. Thermal expansion calculations shall include operating cases where the specified maximum design temperature of either side of the exchanger may occur while the opposite side is at the specified minimum ambient design temperature.
10. If necessary, vendor shall provide expansion bellows for the coolant connections between the bundle and the shell. Expansion bellows shall meet the following minimum requirements:
 - EJ type, design movements and required spring rates shall be determined by Condenser supplier
 - Bellows shall be designed for the following pressure conditions :
 - Atmospheric Pressure internal & Hot Side Design Pressure external
 - Cold Side Design Pressure internal & Full Vacuum external
 - Bellows shall be constructed of Alloy 625 LCF (UNS 06626)
 - Bellows and other components of the EJ shall be designed, fabricated and tested in accordance with Appendix X of ASME B31.3 Code and EJMA standard
 - Bellows shall be designed for at least 200 full coincidental pressure and temperature cycles at design conditions and movements, in accordance with Appendix X of ASME B31.3 Code
 - Bellows shall be single ply construction formed from a tube having longitudinal seams only
 - Internal liners are not permitted
11. Exchanger shall be free draining on the water side and the process side.
12. High point vent nozzle(s), NPS 2 minimum, shall be provided.
13. A steam-out nozzle, NPS 2 minimum, shall be provided at the bottom of the shell at the opposite end from the vent. The nozzle shall be flanged with a gate valve and a blank flange.



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GENERAL NOTES-continued

14. NPS 2 minimum chemical addition nozzles shall be included on the cooling water inlet and outlet nozzles. The diameter of the chemical addition nozzles shall be no less than 1/4 of the cooling water nozzle diameters. The welded plate exchanger supplier shall identify a procedure, including the recommended chemical cleaning agent(s), for chemical cleaning of the cold side of the exchanger, while in place. The welded plate exchanger supplier shall also confirm the size, orientation and quantity of chemical addition nozzles.
15. A blanked off ventilation nozzle on the top of the vessel near the end opposite the manway, NPS 8 minimum, shall be provided.
16. 24 inch minimum ID manways shall be provided at both ends of the horizontal shell plus one on the top of the shell (not on a process inlet nozzle) to facilitate access to all parts of the bundle.
17. Vendor shall provide and describe all access for inspection and repair of the bundle, including the welded ends of the plates as well as the top and bottom edges of the plates.
18. Vendor shall provide field procedures for leak checking and repairs to the bundle.
19. A splash cover shall be provided over the top of the flow entrance to the liquid outlet boot.
20. Access to the inside of the liquid outlet boot shall be provided. If necessary, the splash cover shall included a removable panel providing access to the liquid outlet boot, and corresponding climbing rungs shall be provided inside the liquid outlet boot to permit access to the bottom of the liquid outlet boot.
21. Contractor shall confirm with vendor the maximum permissible nozzle loads. Nozzle loads should be minimized. Contractor to design piping to all process nozzles to obtain these loads or less at all design cases.
22. Shell, bundle, and expansion joints shall be designed for all mechanical and thermal loads that are imposed by the cases defined in Table 1 on Sheet 6.
23. A Helium leak test of the completed heat exchanger shall be performed by vendor prior to shipment in accordance with the requirements of ASME Code, Section V, Article 10. The plates shall be pressurized with a minimum 90% (volume) helium gas at a minimum pressure level of 15 psig (1.0 kg/cm²(g)). The exchanger shall be inclined to provide good flow of helium to a single collection point. Any leakage in excess of 1 x10⁻⁴ cc/sec warrants repair. A leak test shown to be equivalent to this test is acceptable with approval from UOP.
24. The plate bundle internal design pressure is the difference between the cold side design pressure and full vacuum on the hot side. The plate bundle external design pressure is the difference between hot side design pressure and atmospheric pressure on the cold side.
25. Vessel and boot heads shall be 2:1 elliptical heads.
26. NPS 2" Vent / Drain nozzles for shell side and tube side shall be supplied with blind flanges, gaskets flanges, gaskets and bolts / nuts.



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REV

GENERAL NOTES-continued

Case =>	1	2	3	4	5
Normal Operation					
Inlet Temp. °C	52	211	229	178	142
Pressure, kg/cm ² g	-1.0073	2.239	3.449	2.109	0.352
Inlet Flow, kg/hr	181,613	866,362	903,449	2,130	
Vapor Density, kg/m ³	0.1010	10.1166	13.3512	1.6570	0.7176
Vapor Viscosity, cP	0.0067	0.0102	0.0105	0.0138	0.0139
Vapor Specific Heat, kcal/kg-°C	0.3191	0.4575	0.4706	0.5278	0.4836
Vapor Therm. Cond., kcal/hr-m-°C	0.0082	0.0220	0.0233	0.0251	0.0243
Dew Point Temperature, °C	49	211	229	134	108
Liquid Density, kg/m ³	840.4	732.9	714.0	931.0	952.2
Liquid Viscosity, cP	0.5695	0.1039	0.0789	0.2053	0.2590
Liquid Specific Heat, kcal/kg-°C	0.4272	0.5964	0.6238	1.0215	1.0105
Liquid Therm. Cond., kcal/hr-m-°C	0.1018	0.0747	0.0734	0.5891	0.5855
Average Latent Heat, kcal/kg	87.2	74.9	71.5	534.0	516.0
External Pressure	full vacuum, 52 °C	full vacuum, 211 °C	full vacuum, 229 °C	full vacuum, 178 °C	full vacuum, 142 °C
Notes	Continuous; Normal forward flow; Cooling water is flowing at normal rate; Cooling water may be lost at any point during this case	Continuous; Flow in through process inlet nozzles; Cooling water is flowing at normal rate; Cooling water may be lost at any point during this case	Start from Case 2; 15 min duration; Flow in through process inlet nozzles; A second peak event, identical to the first one, may occur some time after the first peak; Between peaks, Case 2 conditions would apply; Cooling water is flowing at normal rate; Cooling water may be lost at any point during this case	Start from normal operation; 2 hour duration; Flow in through vapor outlet nozzles; Cooling water is flowing at normal rate; Cooling water may be lost at any point during this case	Continuous; Flow in through process inlet nozzles; Repeated cycles (1 cycle every 2 years); Cooling water is flowing at normal rate; Cooling water may be lost at any point during this case
Estimated Frequency	Daily - continuous	1 case every 10,000 years	1 case every 10,000 years	1 case per year	1 case every 2 years
Criteria					
Pressure vessel					
Bundle					
Devices for leak-tightness between inlet and outlet of the process side, and connections to the cold side	No damage	Potential damage may occur during this event but shall be repairable	No damage on bundle	No damage	No damage

TABLE 1 - Design Cases

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2	27-Aug-12	MW	LDH				

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Process unit	Item No.		E-2301		
Service of Unit	PREFLASH AND FLASH COLUMNS CONDENSER			No. of units	1
Exchanger Type	Welded Plate		Connected In:	1 Parallel	1 Series
CONFIGURATION					
No. Hot Fluids	1	No. Cold Fluids	1	Total Heat Exchanged	16.52
Fouling Allowance, %		10.0			
CONSTRUCTION PER UNIT					
		Hot Side		Cold Side	
Design Pressure	kg/cm²[g]	4.000		10.5	
Vacuum pressure @ Ref. Temp	kg/cm² [a]	0 @ 52 °C		@	
Design temp. / MDMT	°C	250 / 15		120 / 15	
Flange Rating & Facing		150 RF		150 RF	
Corrosion Allowance	mm	1.5		1.5	
MATERIALS OF CONSTRUCTION					
Components Exposed Only to Hot Fluid			Type 304L SS (Std 3-15)		
Components Exposed to Cold Fluid or Both Fluids			Type 316L SS (Std 3-15)		
NOTES					
1. Design for 105% of flows and duty shown. At 105% design case for flow and duty, the allowable pressure drop may be increased accordingly; the corresponding temperature and pressure at the inlet of the condenser will be 52.6 °C and -1.0066 kg/cm²g. 2. Two hot side process inlet lines of 72" nominal diameter will provide a total of 138,722 kg/hr of flow from the Preflash Column. One hot side process inlet line of 56" nominal diameter will provide 42,891 kg/hr of flow from the Flash Column. These three nozzles will enter the top of the horizontal shell, with the Flash Column nozzle between the two Preflash Column nozzles. 3. The hot side vapor outlet is to two parallel lines, each line diameter is 26". The hot side liquid outlet line diameter connected to the bottom of the boot is 10". 4. The maximum allowable leakage of cooling water into the process is 5 wt ppm based on the hot fluid total fluid quantity. 5. The allowable cooling water velocities are: 1.5 - 2.5 m/s. 6. Water flow may be adjusted to obtain minimum velocity. 7. The cooling water outlet shall be cascaded to the Decomposer Cooler, E-2307. Contractor to confirm flow rate and outlet temperature. 8. Contractor to confirm cold side (cooling water) design pressure and operating pressure. 9. The outlet pressure of this exchanger is a fixed control pressure. The inlet pressure will vary with the actual condenser pressure drop.					

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0	17-May-12	LDH	PFD																																														
1	29-Jun-12	LDH	PFD																																														
REV										SERVICE PREFLASH AND FLASH COLUMNS CONDENSER ITEM NO. E-2301																																							
HOT FLUID PROPERTIES																																																	
Fluid Name CUMENE & CHP				HOT FLUID INLET				HOT FLUID OUTLET																																									
Total Fluid Quantity, kg/h		rel. molecular mass		181613		119.9		181613		119.9																																							
Total Vapor		rel. molecular mass		181613		119.9		2879		102.0																																							
Vapor HCBN				181500		120.2		2768		119.7																																							
Vapor H2																																																	
Vapor H2O				60		18.0		58		18.0																																							
Noncondensibles				53		28.3		53		28.3																																							
Liquid		rel. molecular mass						178734		120.2																																							
Water or Other Liquid		rel. molecular mass																																															
Mole % H2, Vapor		Mole % H2S, Total																																															
Line Sizes, No.		Size, NPS		Note 2				Note 3																																									
Temperature °C				51.7				38.0																																									
Inlet Pressure -1.0073 kg/cm²[g]		Pressure Drop .00680 kg/cm²		Heat Exchanged 16.52 10⁴kcal/h																																													
				VAPOR				TOTAL LIQUID																																									
Pressure	Temp.	Mass Vapor	Enthalpy Normalized	Density	Visc.	Thermal Cond.	Specific Heat	Density	Visc.	Thermal Cond.	Specific Heat	Surface Tension	Critical Press.	Critical Temp																																			
kg/cm²[g]	°C		kcal/kg	kg/m³	cP	kcal/(h.m.°C)	kcal/(kg.°C)	kg/m³	cP	kcal/(h.m.°C)	kcal/(kg.°C)	dyn/cm	kg/cm² [a]	°C																																			
-1.0066	65.5	1.0000	100.00	.09931	0.007	.00966	0.34																																										
-1.0066	57.4	1.0000	97.31	.1025	0.007	.00898	0.33																																										
-1.0066	49.1	1.0000	94.62	.1041	0.007	.00826	0.32	839.9	0.566	.102	0.43	25.5	32.8	358																																			
-1.0066	48.9	0.8450	81.10	.1041	0.007	.00825	0.32	838.7	0.557	.102	0.43	25.3	32.7	358																																			
-1.0066	48.8	0.5340	54.07	.1041	0.007	.00825	0.32	838.3	0.554	.102	0.43	25.2	32.7	358																																			
-1.0066	48.8	0.3785	40.55	.1041	0.007	.00825	0.32	838.3	0.554	.102	0.43	25.2	32.7	358																																			
-1.0066	48.6	0.2234	27.03	.1041	0.007	.00826	0.32	838.4	0.555	.102	0.43	25.2	32.7	358																																			
-1.0066	48.0	0.0709	13.52	.1009	0.007	.00832	0.32	838.9	0.558	.102	0.43	25.3	32.7	358																																			
-1.0066	38.0	0.0048	3.56	.06888	0.008	.00950	0.31	847.3	0.627	.104	0.42	26.3	32.8	358																																			
-1.0066	29.9	0.0022	0.00	.05286	0.010	.0108	0.31	854.0	0.694	.105	0.41	27.2	32.8	358																																			
-1.0073	65.5	1.0000	100.00	.09611	0.007	.00966	0.34																																										
-1.0073	57.2	1.0000	97.22	.09931	0.007	.00895	0.33																																										
-1.0073	48.6	1.0000	94.44	.1009	0.007	.00821	0.32	840.4	0.570	.102	0.43	25.5	32.8	358																																			
-1.0073	48.4	0.8455	80.94	.1009	0.007	.00819	0.32	839.2	0.560	.102	0.43	25.4	32.7	358																																			
-1.0073	48.3	0.5352	53.96	.1009	0.007	.00819	0.32	838.8	0.558	.102	0.43	25.3	32.7	358																																			
-1.0073	48.2	0.3801	40.47	.1009	0.007	.00821	0.32	838.8	0.558	.102	0.43	25.3	32.7	358																																			
-1.0073	48.1	0.2254	26.98	.1009	0.007	.00821	0.32	838.8	0.558	.102	0.43	25.3	32.7	358																																			
-1.0073	47.4	0.0731	13.49	.09771	0.007	.00826	0.32	839.3	0.562	.102	0.43	25.3	32.7	358																																			
-1.0073	38.0	0.0051	3.59	.06888	0.008	.00938	0.31	847.3	0.627	.104	0.42	26.3	32.8	358																																			
-1.0073	29.9	0.0023	0.00	.05126	0.010	.0106	0.31	854.0	0.694	.105	0.41	27.2	32.8	358																																			

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